

Class #24 - Change of Basis

1. Given an n -dimensional nonzero vector space V (we automatically have a _____), each $x \in V$ is identified _____ by _____, i.e., $[x]_B$ serves as a "name" for x that lists the weights used to build x as a linear combination of the basis vectors in B .)

2. In some applications, a problem is described initially using a basis B , but the problem's solution is aided by changing B to a new basis S .

Question: How are $[x]_B$ and $[x]_S$ are related?

3. Example: Consider two bases $B = \{b_1, b_2\}$ and $S = \{s_1, s_2\}$ for a vector space V , such that

$$b_1 = 4s_1 + s_2 \quad \text{and} \quad b_2 = -6s_1 + s_2.$$

Suppose that $x = 3b_1 + b_2$, i.e., _____ = $\begin{bmatrix} 3 \\ 1 \end{bmatrix}$. Find $[x]_S$.

4. Let $B = \{b_1, \dots, b_n\}$ and $S = \{s_1, \dots, s_n\}$ be bases of a vector space V . Then there is an _____ matrix $M_{S \leftarrow B}$ such that

$$[x]_S = M_{S \leftarrow B} [x]_B.$$

The matrix $M_{S \leftarrow B}$ is called the *change-of-coordinates matrix* from _____ to _____.

The columns of $M_{S \leftarrow B}$ are the S -coordinate vectors of the vectors in the basis B , i.e.,

$$M_{S \leftarrow B} = [[b_1]_S \cdots [b_n]_S].$$

5. Note: The columns of $M_{S \leftarrow B}$ are _____:

Hence, the matrix $M_{S \leftarrow B}$ is invertible, and

$$(M_{S \leftarrow B})^{-1} = \text{_____}.$$

6. Example: Let $b_1 = \begin{bmatrix} -9 \\ 1 \end{bmatrix}$, $b_2 = \begin{bmatrix} -5 \\ -1 \end{bmatrix}$, $s_1 = \begin{bmatrix} 1 \\ -4 \end{bmatrix}$, $s_2 = \begin{bmatrix} 3 \\ -5 \end{bmatrix}$, and consider the bases for $\mathbf{R}^2$ given by $B = \{b_1, b_2\}$ and $S = \{s_1, s_2\}$.

(a) Find the change-of-coordinates matrix from B to S .

(b) Find the change-of-coordinates matrix from S to B .